

A counterexample to TLMC conjecture 00000002320

AI-assisted mathematical exploration

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Abstract

The conjecture places the probabilities of generating all finite groups by two uniform random elements in the interval $[1/4, 1]$. The elementary abelian group $(\mathbb{Z}/2\mathbb{Z})^3$ has no generating pair, so its probability is zero. We describe the standard counting definition, the counterexample, and its correspondence with the accompanying Lean source.

Target and conventions

Conjecture 00000002320 of The Last Math Competition, at the fixed revision identified in reference [1], asserts that the spectrum of generating-pair probabilities over *all finite groups* is a dense subsequence of $[1/4, 1]$. It also makes a separate assertion about $\mathrm{PSL}(2, p)$. We disprove the necessary consequence

$$P(G) \geq \frac{1}{4} \quad \text{for every finite group } G.$$

The source does not restrict the quantifier to simple groups or to groups that can be generated by two elements.

For a finite group G , let

$$N_2(G) = |\{(a, b) \in G \times G : \langle a, b \rangle = G\}|, \quad P(G) = \frac{N_2(G)}{|G|^2}.$$

Thus the two elements are sampled independently and uniformly, with replacement; pairs are ordered. This is the standard definition of the two-generator probability; see Pak [2, p. 3] for the general k -tuple definition.

Proposition 1. *Let $G = (\mathbb{Z}/2\mathbb{Z})^3$. Then $|G| = 8$ and $P(G) = 0$. Consequently the spectrum asserted in conjecture 00000002320 is not contained in $[1/4, 1]$.*

Proof. Write the group operation additively. For any $a, b \in G$, consider

$$H_{a,b} = \{0, a, b, a + b\}.$$

The group is abelian and every element satisfies $x + x = 0$. It follows that $H_{a,b}$ contains zero and is closed under addition and inverses, so it is a subgroup containing a and b . Therefore

$$\langle a, b \rangle \subseteq H_{a,b}, \quad |\langle a, b \rangle| \leq 4 < 8 = |G|.$$

No pair generates G . There are $8^2 = 64$ ordered pairs in total and zero generating pairs, giving

$$P(G) = \frac{0}{64} = 0 < \frac{1}{4}.$$

This contradicts the required interval containment. □

Lean correspondence and verification scope

The accompanying file `Counterexample2320.lean` uses the actual mathlib group `Multiplicative (ZMod 2 × ZMod 2 × ZMod 2)`. The multiplicative type tag only changes the notation for the operation. The generating relation is mathlib's `Subgroup.closure` of the two-element set, equal to the top subgroup.

The declarations `fourSubgroup` and `pair_closure_le_four` construct a subgroup with carrier $\{1, a, b, ab\}$ and contain the generated subgroup in it. The finite theorem `four_misses_element` supplies an element outside this subgroup for every pair.

Consequently `no_pair_generates` proves that no generated subgroup is top. The definition `generatingPairs` filters the full finite set of ordered pairs by this exact generation predicate, and `generatingPairProbability` is its cardinality divided by $|G|^2$. The terminal declarations are `generatingPairProbability_zero` and `conjectureInterval_false`.

The probability is represented exactly as a rational number. Its usual embedding into the reals preserves the value zero and the comparison with $1/4$. Finite checks use kernel-checked `decide`; no `native_decide` is used. Compilation, axiom auditing, and replay evidence are recorded separately in the delivered validation logs.

The argument refutes one necessary consequence of the conjecture. It does not claim a formal proof concerning the separate $\text{PSL}(2, p)$ limit. The fact about elementary abelian groups is standard, and no novelty in group theory is claimed. A bounded check of the repository's 100 pull request titles and bodies at 2026-09-20 17:13 UTC found no matching submission identifier; this is not a claim of worldwide priority.

References

- [1] The Last Math Competition, *Conjecture 00000002320*, revision 95acb520ec5607c826b8a997b1ef2fc82d6f7c57. Fixed upstream source.
- [2] Igor Pak, *On probability of generating a finite group*, December 30, 1999, especially p. 3. <https://www.math.ucla.edu/~pak/papers/sim.pdf>.